NOT MEASUREMENT SENSITIVE

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SUPERSEDING
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MILITARY SPECIFICATION

# LUBRICANT, SOLID FILM, AIR CURED (CORROSION INHIBITING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of an air-curing solid film lubricant (see 6.1) identified by Military Symbol SFD. The lubricant provides both lubrication and corrosion protection and can be applied by dipping, by brushing, or by spraying from gas-pressured (aerosol) cans.

1.2 Classifications. This solid film lubricant shall be furnished in the following types, forms, and colors as specified (see 6.2):

| | |
|---|---|
| Type I - | An air dry lubricant with an endurance life of 120 minutes and cure time of 18 hours. |
| Form 1 | - Bulk dispersion |
| Color 1 | - Natural product color. |
| Color 2 | - Black color No. 37038, but not lighter than Gray No. 36076 from FED-STD-595 |
| Form 2 | - Aerosol propelled |
| Color 1 | - Natural product color. |
| Color 2 | - Black color No. 37038, but not lighter than Gray No. 36076 from FED-STD-595. |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA MOBILITY TECH CENTER - BELVOIR, ATTN AMSTA-RBES, GRIDLEY RD, STE 128, FT BELVOIR, VA 22060-5843 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Type II - An air dry lubricant with low volatile organic compound (VOC) content and an endurance life of 90 minutes and cure time of 24 hours.

Color 1 - Natural product color.
Color 2 - Black color No. 37038, but not lighter than Gray No. 36076 from FED-STD-595.

## 2. APPLICABLE DOCUMENTS

### 2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| QQ-A-250 | - | Aluminum Alloy and Aluminum Alloy Plate Sheet. |
| QQ-A-250/5 | - | Aluminum Alloy, Alclad 2024, Plate and Sheet. |
| TT-N-95 | - | Naphtha, Aliphatic. |
| VV-D-1078 | - | Damping Fluid, Silicone Base (Dimethyl Polysiloxane). |
| VV-L-800 | - | Lubricating Oil, General Purpose, Preservative (Water-Displacing, Low Temperature). |
| PPP-C-96 | - | Cans, Metal, 28 Gage and Lighter. |

MILITARY

| | | |
|---|---|---|
| MIL-C-372 | - | Cleaning Compound, Solvent (For Bore of Small Arms and Automatic Aircraft Weapons). |
| MIL-L-6082 | - | Lubricating Oil, Aircraft Reciprocating Engine (Piston). |
| MIL-L-7808 | - | Lubricating Oil, Aircraft Turbine Engines, Synthetic Base. |
| MIL-A-8625 | - | Anodic Coatings for Aluminum and Aluminum Alloys. |
| MIL-L-14107 | - | Lubricating Oil, Low Temperatures, Weapons. |
| DOD-P-16232 | - | Phosphate Coatings, Heavy, Manganese or Zinc Base (For Ferrous Metals). |
| MIL-L-46000 | - | Lubricating Oil, Semi-Fluid (Automatic Weapons). |
| MIL-H-46170 | - | Hydraulic Fluid, Rust Inhibited, Fire. Resistant, Synthetic Hydrocarbon Base. |



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MIL-L-63460 - Lubricant, Cleaner and Preservative for Weapons.
MIL-T-83133 - Turbine Fuel, Aviation, Kerosene Types.

STANDARDS

FEDERAL

FED-STD-595 - Colors Used in Government Procurement.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-290 - Packaging, Packing, and Marking of Petroleum and Related Products.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from. STDZN DCMNT ORDER DESK, BLDG 4D, 700 ROBBINS AVE, PHILADELPHIA PA 19111-5094.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 108 - Steel Bars, Carbon, Cold Finish, Standard Quality.
B 117 - Salt Spray (Fog) Test.
B 244 - Measurement of Thickness of Anodic Coatings on Aluminum and Other Non-Conductive Coatings on Non-Magnetic Basis Metals with Eddy Current Instruments.
B 499 - Measurement of Coating Thickness by the Magnetic Method: Non Magnetic Coatings in Magnetic Basis Metals.
D 1186 - Nondestructive Measurement of Dry Film Thickness of Nonmetallic Coatings Applied to a Ferrous Base.
D 1193 - Reagent Water.
D 1475 - Density of Paint, Varnish, Laquer, and Related Products.
D 1400 - Non Destructive Measurement of Dry Film Thickness of Non-conductive Coating applied to a Non-Ferrous Metal Base.
D 2369 - Volatile Content of Coatings.
D 2510 - Adhesion of Solid Film Lubricants.
D 2511 - Thermal Shock Sensitivity of Solid Film Lubricants.



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D 2625 - Endurance (Wear) Life and Load Carrying Capacity of Solid Film Lubricants (Falex Method).
D 2832 - Determining Non Volatile Content of Paint and Paint Materials.
D 3960 - Standard Practice for Determining Volatile Organic Compound (VOC) Content of Paints and Related Coatings.
D 4017 - Water in Paints and Paint Materials by Karl Fischer Method.
D 4457 - Determination of Dichloromethane and 1,1,1-trichloroethane in Paints and Coatings by Direct Injection into a Gas Chromatograph.
F 22 - Standard Test Method for Hydrophobic Surface Films By the Water-Break Test.

(Application for copies should be addressed to: AMERCN SCTY FOR TEST & MTRLS, 1916 RACE STRET, PHILADELPHIA PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3 1 Qualification. Solid film lubricants furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for award of bids (see 4.5.1 and 6.3). Any change in the formulation of a qualified product shall necessitate its requalification.

3.2 Materials. The lubricant shall be of the following composition:

a. A suitable, fast-drying vehicle such that the applied film is dry to the touch in a maximum of 30 minutes at 25 ±2 °C and shall cure fully in a maximum of 18 hours at 25 ±2 °C for type I and dry to the touch in 90 minutes at 25 ±2 °C and fully cured in a maximum of 24 hours at 25 ±2 °C for type II

b. A suitable lubricative pigment or mixture of lubricative pigments.

c. Additives, if necessary

d. For lubricant in gas-pressured cans only, a pressure-producing agent to expel the contents of the container, which contains no ozone depleting substances.



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3.3 Film appearance and thickness. The bonded solid film lubricant, when examined as specified in 4.6, table I shall appear uniform in color and shall be smooth, free from any cracks, sags, foreign matter, grit, rough particles, or separation of ingredients. The thickness of the cured film shall be between 0.008mm and 0.013mm with no single reading less than 0.005mm or greater than 0.018mm.

3.4 Film adhesion. The lubricant, when applied to anodized aluminum panels, cured to a film thickness of 0.008 to 0.013mm and tested with tape in accordance with table II (ASTM D 2510) shall not be lifted to expose any bare surface. A uniform deposit of powdery material clinging to the tape shall not be considered cause for rejection.

3.5 Endurance life. The lubricant, when applied to phosphated steel specimens and cured to a film thickness of 0.008 to 0.013 mm, shall have an average Falex endurance life of not less than 120 minutes under a gage load of 1000 lbf (4480 N) for type I and an average of not less than 90 minutes under a gauge load of 1000 lbf (4480 N) for type II. The test shall be performed in accordance with table II (ASTM D 2625). No test result shall be less than 100 minutes for type I and 75 minutes for type II.

3.6 Load-carrying capacity. The lubricant, when applied to phosphated steel specimens and cured to a film thickness of 0.008 to 0.013 mm shall have an average Falex load-carrying capacity of not less than 2500 lbf (11,200 N) with no single test result less than 2000 lbf (8950 N) The test shall be performed in accordance with table II (ASTM D 2625).

3.7 Corrosion protection (salt spray). The lubricant, when applied to phosphated steel panels and cured to a film thickness of 0.008 to 0.013 mm, shall show no more than three rust spots per panel, none of which exceeds one millimeter in length, width, or diameter after 100 hours exposure to a 5-percent salt spray solution in a salt fog cabinet. The test shall be performed in accordance with table II (ASTM B 117).

3.8 Thermal shock sensitivity. The lubricant, when applied to aluminum panels and cured to a film thickness of 0.008 to 0.013 mm, shall not flake, crack or soften and shall pass the film adhesion requirement (3.4). The test shall be performed in accordance with the table II (ASTM D 2511), except that the high temperature shall be 191 ±3 °C.

3.9 Fluid resistance. The lubricant, when applied to aluminum panels and cured to film thickness of 0.008 to 0.013 mm, shall not flake or peel and shall pass the film adhesion requirement (3.4) after being immersed in the test fluids specified in table I. The test shall be performed in accordance with table II (ASTM D 2510).



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TABLE I. Immersion fluids.

| Fluid | Specification |
|---|---|
| Silicone Damping Fluid<br>Reagent Water | VV-D-1078<br>ASTM D 1193 |
| Cleaning Compound, Solvent (for Bore of Small Arms and Automatic Weapons) | MIL-C-372 |
| Lubricating Oil, General Purpose, Preservative (Water Displacing, Low Temperature)<br>Hydraulic Fluid, Synthetic Hydrocarbon | VV-L-800<br>MIL-H-46170 |
| Turbine Fuel, Aviation, Kerosene Types | MIL-T-83133, grade JP-8 |
| Lubricating Oil, Aircraft | MIL-L-6082, grade 1100 |
| Lubricating Oil, Aircraft Turbine | MIL-L-7808 |
| Lubricating Oil, Semi-Fluid, (Automatic Weapons) | MIL-L-46000 |
| Lubricant, Cleaner and Preservative for Weapons<br>Lubricating Oil, Low Temperature, Weapons | MIL-L-63460<br>MIL-L-14107 |

3.10 Spray pattern and duration (lubricant in gas-pressured containers only). The spray pattern shall be a minimum of 38 mm wide and shall be uniform in color, smooth, and free from bubbles The spray shall be effective for a minimum of 290 seconds. (The effective spray is one that carries the pigment together with the resin required for bonding.) The test shall be performed in accordance with 4.6.2.

3.11 Total solids

3.11.1 Lubricant in non-pressured cans Lubricant supplied in non-pressurized cans shall contain a minimum of 24 percent by weight of total solids. The test shall be performed in accordance with 4.6.3.1.

3.11.2 Lubricant in gas-pressured cans. Each gas-pressurized can shall contain a minimum of 30 grams of total solids. The test shall be performed in accordance with 4.6.3.2.

3.12 Graphite and powdered metals. The lubricant shall contain no graphite or powdered metals in accordance with 4.6.4.

3.13 Storage stability The lubricant, after being stored for 365 ±7 days at 25 ±3 °C shall be readily dispersed by moderate shaking or stirring and shall meet all requirements in section 3 except total solids (3.11) and graphite and powered metals (3.12). The test shall be performed in accordance with 4.6.5.



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3.14 Volatile organic compound content. The lubricant (type II only) shall contain not more than 250g/L of volatile organic compound content when tested as specified in 4.6.6.

3.15 Color. When visually examined, the black color option shall closely match Black No. 37038, but shall not be lighter than Gray No. 36076 from FED-STD-595.

3.16 Workmanship. The lubricant ingredients shall be processed in such a manner as to insure that the lubricant will produce a bonded solid film which shall, upon visual examination, appear uniform in color, smooth, free from cracks, scratches, blisters, foreign matter, grit, rough particles, bubbles, pin holes, runs, sags, or other surface imperfections. The lubricant film shall show no evidence of separation of ingredient materials.

3.17 Application to end item. All end items shall have the lubricant applied in accordance with the appendix of this document unless otherwise specified in the purchase order, contract, end item drawing, or end item specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract and unless disapproved by the Government, the contractor's own or any other facilities suitable for the performance of the inspection requirements specified herein, may be used. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot.

4.2 1 Bulk lot. An indefinite number of unit containers of identical size and type offered for acceptance and filled with a homogeneous mixture of lubricant, offered for acceptance in a single isolated container; or filled with homogeneous mixture of lubricant manufactured in a single plant run (not exceeding 24 hours); through the same processing equipment, with no change in the ingredient materials.

4.2.2 Packaged lot. An indefinite number of unit containers of identical size and type offered for acceptance and filled with a homogeneous mixture of lubricant from a single isolated container; or filled with a homogeneous mixture of lubricant manufactured in a single plant run (not exceeding 24 hours); through the same processing equipment, with no change in the ingredient materials.



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4.3 Sampling.

4.3.1 Sampling for examination of filled containers. Take a random sample of filled containers from each lot in accordance with MIL-STD-105 at inspection level II.

4.4 Examination of filled containers. Examine samples taken in accordance with 4.3.1 for compliance with MIL-STD-290 with regard to fill, closure, sealing, leakage, packaging, packing and marking of the containers. Reject any container having one or more defects and under the required fill. If the number of defective or underfilled containers exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, reject the lot represented by the sample.

4.5 Classification of tests. Tests are classified as follows:

a. Qualification tests.
b. Quality conformance tests.

4.5.1 Qualification tests Qualification tests consist of tests for all of the requirements specified in section 3.

4 5.2 Quality conformance tests Quality conformance tests consist of tests for the following requirements specified in section 3:

a. Endurance life (3.5).
b. Corrosion protection (salt-spray) (3.7).
c Total solids (3.11).
d. Film adhesion (3.4).

4.6 Test methods Perform tests in accordance with table II and 4.6.1 through 4.6.5 on specimens which have been coated with the solid film lubricant and cured at 25 ±2 °C for a minimum of 24 hours. The specimens prepared for tests shall be prepared by spraying



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TABLE II. Test methods.

| TEST | METHOD NUMBER (ASTM) | REQUIREMENT |
|---|---|---|
| Film Appearance & Thickness aluminum | D 14400 or B 244 | 3.3 |
| steel | D 1186 or B 499 | 3.3 |
| Film Adhesion | D 2510 1/ Procedure A | 3.4 |
| Endurance Life | D 2625 1/ Procedure A | 3.5 |
| Load Carrying Capacity | D 2625 1/ Procedure B | 3.6 |
| Corrosion Protection (salt spray) | B 117 1/,2/ | 3.7 |
| Thermal Shock Sensitivity | D 2511 1/,3/ | 3.8 |
| Fluid resistance | D 2510 1/,4/ Procedure C | 3.9 |
| Volatile organic content | D 3960<br>D 2369<br>D 1475<br>D 4017 | 3.14 |

1/ For surface cleaning, use only non-ODS cleaning solvents in these test method procedures. A recommended substitute is Aliphatic Naphtha (TT-N-95). Any environmentally compliant cleaner may be used which sufficiently cleans surfaces to pass ASTM F 22 but does not cause damage to the surface (i.e., Hydrogen Embrittlement, etc.)

2/ Determine the solid film lubricant coating thickness in accordance with ASTM B 1186 or D 499.

3/ Perform this test with the following exceptions:
Use anodized aluminum panels as prescribed in ASTM D 2510 and an oven temperature of 191 ±3°C.

4/ Make two parallel scratches one hour after drying.

4.6.1 Preparation of test panels.

4.6.1.1 Preparation of and application to test panels for solid film lubricant (aluminum). The panels shall be made from aluminum alloy conforming to QQ-A-250/5, anodized to conform to MIL-A-8625, type II sulfuric acid anodize, and measuring approximately 0.05 cm by 7.6 cm by 15.2 cm. The panels shall be pre-cleaned with aliphatic naphtha conforming to TT-N-95 or any



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EPA compliant cleaner that sufficiently cleans the surfaces to pass ASTM F 22. Application of the lubricant shall be performed in a well-ventilated area or hood where no flames or ignition sources are present. Only one side of each panel shall be fully coated, except for two of the anodized aluminum panels which shall have the lubricant applied to a 2.54 cm wide strip to enable measurement of the film thickness. A spray application technique shall be used to coat the panels for the tests specified herein The solid film lubricant thickness, after cure, shall be 0.008 to 0.013 mm Three coats shall be the maximum number required to obtain the desired film thickness. After the final coat has been applied, the coated specimens shall be allowed to air dry for 18 (type I) or 24 (type II) hours for complete cure. At least two test panel specimens shall be used in each test method. A total of 22 aluminum panels are required for testing in accordance with performance requirements of this specification.

4.6.1.2 Preparation of and application to test panels for solid film lubricant (steel). The panels shall be steel panels approximately 0.32 cm by 7.6 cm by 15.2 cm conforming to ASTM A 108. The panels shall be pre-cleaned with Aliphatic Naphtha conforming to TT-N-95 or any EPA compliant cleaner that sufficiently cleans surfaces to pass ASTM F 22 The panels shall have both faces and all edges grit-blasted with 180-220 grit aluminum oxide for a surface finish of 20-40 Ra. Phosphate the panels (weight should be 11 - 22 $g/m^2$) in accordance with DOD-P-16232, using ASTM B 499 or D 1186 to measure the thickness of the phosphated panels The steel panels shall be sprayed to the same thickness as specified for the aluminum panels. The coated specimens shall be allowed to air dry for 18 (type I) or 24 (type II) hours for complete cure. At least two test panel specimens shall be used in each method. A total of two steel panels and six sets of pins and vee blocks are required for testing in accordance with performance requirements of this specification.

4.6.2 Spray pattern and duration (material in gas-pressurized cans only). Select a new, unused spray container. The temperature of the spray container and the ambient air shall be 25 ±2 °C. In a vertical position, fasten either a sheet of white paper on which two parallel lines have been drawn 38 mm apart, or a sheet of rectangular coordinate graph paper containing lines 38 mm apart. Position the spray container 25 ±1.25 cm from the paper. Open the valve fully and, with a stop watch, determine the time required to exhaust the container. Measure the width of the spray pattern using the parallel lines on the sheet of paper. Agitate the can frequently during the spray procedure.

4.6.3 Total solids.

4.6.3.1 Bulk material Stir the lubricant thoroughly to provide a uniform dispersion. Weigh 5 ±0.5 grams of the lubricant into a weighing dish with a diameter of approximately



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64 mm (Fisher Scientific Co. Catalog No. 7-732 or equal). Place the dish and contents in a force-draft oven maintained at 49 ±3 °C for 18 ±1 hour. Remove the dish and contents from the oven, place them in a desiccator, and allow them to cool to 25 ±3 °C. Remove the dish and contents from the desiccator and weigh them. Repeat the procedure to constant weight. Calculate the percent by weight of the solid material in the fluid lubricant from the formula:

$$\text{Percent total solids} = \frac{\text{Weight of solid materials (grams)}}{\text{Weight of sample (grams)}} \times 100$$

4.6.3.2 Lubricant in gas-pressurized cans. Select a new, unused spray container for the total solids test (3.11.2). Place it in an upright position in a solid carbon dioxide cabinet (dry ice cabinet) overnight to reduce the internal pressure. Remove the container from the cold cabinet and immediately pierce a small hole in the top of the container with a sharp punch. Permit the container to warm to 25 ±3 °C in an upright position, in a well-ventilated area. When all the gas has escaped from the container, remove the top of the container with a can opener. Transfer quantitatively the contents of the container to a 600-ml beaker previously weighed to the nearest 0.1 gram. Rinse the container twice with an appropriate solvent (as recommended by the manufacturer) or distilled water for water based coating, and add the rinsings to the 600-ml beaker. Remove the agitator (usually a glass marble or a small steel ball) from the beaker, rinse it with methylene chloride and return it to the empty container. Add the methylene chloride rinsing to the 600 ml beaker. Place the beaker in an explosion-proof oven operated at 49 ±3 °C. Permit the 600-ml beaker to remain in the oven overnight. Remove the beaker, permit it to cool to 25 ±3 °C and weigh to a constant weight. Calculate the weight of total solids in grams from the following formula:

Weight beaker and total solids (in grams) - weight beaker (in grams) = Weight total solids (in grams)

4.6.4 Graphite and powdered metals. The manufacturer shall submit a notarized certification to the qualifying activity, signed by a responsible officer in its management, attesting that no graphite or powdered metals are present in the product furnished under this specification.

4.6.5 Storage stability.

4.6.5.1 Lubrication in non-pressured cans. Fill a one-quart (0.946 liter) can conforming to PPP-C-96, type V, class 2 with thoroughly-mixed lubricant to within approximately 6 mm of the top. Seal the can tightly and place it in storage at 25 ±3 °C for 365 ±7 days. At the end of the storage period, test the lubricant for conformity to all the requirements of section 3 except total solids (3.11) and Graphite and Powdered Metals (3.12).



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4.6.5.2 Lubricant in gas-pressured cans. Select two new, unused spray cans and place them in storage at 25 ±3 °C for 365 ±7 days. At the end of the storage period, test the lubricant for conformity to all of the requirements of section 3 except total solids (3.11).

4.6.6 Volatile organic compound content. The VOC content shall be determined in accordance with ASTM D 3960; the non-volatile content shall be determined in accordance with ASTM D 2832; the Water content shall be determined in accordance with ASTM D 4017, the density shall be determined in accordance with ASTM D 1475; and exempt solvents determined in accordance with ASTM D 4457. The VOC content calculation shall be performed in accordance with ASTM D 3960-93.

4.6.7 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of MIL-STD-290 (see 5.1).

5. PACKAGING

5.1 Preservation-packaging, packing, and marking.

5.1.1 Lubricant in non-pressured cans. Unless otherwise specified in the contract (6.2), packaging, packing, and marking shall be level A, B or commercial, as specified in accordance with MIL-STD-290. In addition, each unit package, intermediate package, and external packaged shall be marked as follows:

CAUTION: FLAMMABLE. Use only in a well-ventilated area or in a hood where no flames or other ignition sources are present HARMFUL IF INHALED.

WARNING: Do not use this lubricant in food-processing or food-handling equipment on surfaces that may contact food. Do not allow the lubricant to contaminate foodstuffs.

5.1.2 Lubricant in gas-pressurized cans. Unless otherwise specified in the contract (6.2), packaging, packing, and marking shall be level A, B, or commercial, as specified in accordance with MIL-STD-290. The lubricant shall be packaged in 12 or 16 ounce cans conforming to PPP-C-96, type IX, class 2. Each can shall contain a stirring device such as a marble or a steel ball. In addition, each unit package, intermediate package, and external package shall be marked as follows:

CAUTION: FLAMMABLE. Use only in a well-ventilated area or in a hood where no flames or other ignition sources are present (type I only). HARMFUL IF INHALED



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WARNING: Do not use this lubricant in food-processing or food-handling equipment on surfaces that may contact food. Do not allow the lubricant to contaminate foodstuffs.

WARNING PRESSURIZED CONTAINER: Do not store at a temperature above 49 °C. Keep away from direct sunlight, radiators, stoves, hot water or other heat sources. Do not puncture this can nor place it in an incinerator.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The solid film lubricant covered by this specification is intended for use on aluminum, aluminum alloys, copper and copper alloys, steel, stainless steel, titanium, and chromium and nickel bearing surfaces. It is useful under the following conditions:

a. Where conventional lubricants are difficult to apply or retain.

b. Where dust and dirt contamination on lubricated surfaces is deleterious.

c. Where temperatures may range from -67 to +93°C.

d. In mechanisms operated at infrequent intervals.

e. In mechanisms that are lubricated for the life of the mechanism.

f. Where long-term corrosion protection is required.

g. Where a dull dark gray or black non-reflective surface is required.

h. Where a sacrificial lubricant is necessary to carry extremely heavy loads developed in the initial start-up of heavily loaded mechanism designed for fluid lubrication.

i. In operations consisting of reciprocating motion, loaded to 15 psi (103 kPa) or less, where contamination with conventional fluid lubricants is probable.

j. To touch up worn surfaces originally coated with lubricant conforming to MIL-L-46010.



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k. For sliding motion applications such as plain and spherical bearings, flap tracks, hinges, threads, and cam surfaces.

The lubricant should not be used under the following conditions:

a. In operations consisting of rotary motion above 100 rpm under heavy loads where the possibility of conventional fluid lubricant contamination exists. The cured lubricant film is highly resistant to conventional fluid lubricants, but the high fluid pressures developed in heavily loaded sleeve type bearings drastically reduces the wear life provided by the solid film lubricant film.

b. On bearings containing rolling elements.

This lubricant can be expected to provide corrosion protection for five years in indoor storage and approximately two years protection in outdoor storage when the lubricant is applied over phosphated steel to a thickness of 0.013 mm. Where maximum corrosion protection of steel is desired, the lubricant should be applied over phosphated steel to a thickness of 0.025 mm. This heavier coating can be expected to provide outdoor corrosion protection for approximately four years.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Quantity of lubricant required.
c. Type of lubricant, form of container (pressurized or non-pressurized cans), and color required (see 5.1).
e. Level of packaging and packing (see 5.1).
f Special marking, if required (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractor is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the US Army Mobility Technology Center - Belvoir, ATTN AMSTA-RBF, 10115 Gridley Rd., Suite 128, Fort Belvoir, VA 22060-5843, and information pertaining to qualification of products may be obtained from that activity.



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6.4 Subject term (key word) listing.

Lubricant
Solid film
Dry lubrication
Solid lubrication
Low VOC, corrosion protection.

6.5 Units of measure. In sections of this specification where U.S. customary units are given first with metric units in parenthesis, the U.S. customary units should be considered as the standard. Metric units have been rounded off and are not, in many instances, precise equivalents of the U.S. units.

6.6 Disposability. This product contains lead and thus should be disposed of as a Hazardous Waste in accordance with all Resource Conservation Recovery Act (RCRA) regulations as well as all state and local regulations.

6.7 End item testing. When possible, it is recommended that film thickness and adhesion be tested when the lubricant has been applied to an end item. Testing with laboratory coupons and test panels may not fully support lubricant performance when applied to an actual end item.

6.8 Changes to previous issue: This specification includes the following major changes:

Cover Page:
a. Added two color classifications.
b. Added low volatile organic content (VOC) type II lubricant.

Section 2:
a. Deleted obsolete documents and added new documents as required.

Section 3:
a. Added requirements for type II lubricant to include maximum volatile organic content (VOC), endurance life, and load carrying capacity.
b. Added black color requirements for Color 2.

Section 4:
a. Eliminated use of 1,1,1-trichloroethane as degreaser for precleaning test panels.
b. Added test method for determining VOC content of type II lubricant.

Section 6:
a. Added Classification Changes.



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Appendix:
a. Changed appendix to mandatory part of specification unless otherwise specified in contract or purchase order.
b. Eliminated use of 1,1,1-trichloroethane as degreaser for precleaning surfaces.

Custodians:
Army - ME
Navy - AS
Air Force - 68

Preparing activity:
Army - ME

Project 9150-1097

Review activities:
Army - AT, AV, MD, MI
Navy - OS
DLA - GS, PS



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# INSTRUCTIONS FOR APPLYING LUBRICANT,

# SOLID-FILM: AIR-CURED (CORROSION-INHIBITING)

## 10. SCOPE

10.1 Scope. This appendix covers in detail the surface pretreatment, temperature, and time required to cure the solid film lubricant when it is applied over the bearing surfaces of various metals. This appendix is a mandatory part of the specification unless otherwise specified in the contract, purchase order, end item drawing or end item specification. This appendix is NOT mandatory for fastener hardware unless otherwise specified.

## 20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Specifications and standards. The following specifications standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| QQ-P-35 | - | Passivation Treatments for Austenitic, Ferritic, and Martensitic Corrosion-Resisting Steel (Fastening Devices). |

MILITARY

| | | |
|---|---|---|
| MIL-F-495 | - | Finish, Chemical, Black, for Copper Alloys. |
| MIL-M-3171 | - | Magnesium Alloy, Process for Pretreatment and Prevention of Corrosion on. |
| MIL-M-45202 | - | Magnesium Alloys, Anodic Treatment of. |

## 30. REQUIREMENTS

30.1 General application instructions for all metals. Do not touch the pretreated surfaces with the fingers. Stir the lubricant until thoroughly mixed, using a low shear mixing blade. A mechanical paint shaker may be used for type I, but is not recommended for type II as excessive foaming with waterborne products may occur. Minor viscosity adjustments to type II may



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be made by adding deionized water (1 megohm/cm or greater) DO NOT USE ORDINARY TAP WATER. Apply the lubricant by brushing, dipping, or spraying to a nominal film thickness of 0.01 mm with no reading less than 0.005 mm or greater than 0.018 mm and permit the coated parts to air dry. The application of the coating to parts shall be as specified in 30.2 through 30.7 unless otherwise specified in the contract or purchase order.

30.2 Application on aluminum and aluminum alloys. Preclean the surfaces to be coated with Aliphatic Naphtha conforming to TT-N-95 or any EPA compliant cleaner that sufficiently cleans surfaces to pass ASTM F 22. Anodize and seal the surface in accordance with MIL-A-8625, types I, II, or III, class 1.

30.3 Application on copper and copper alloys. Preclean the surfaces to be coated with Aliphatic Naphtha conforming to TT-N-95 or any EPA compliant cleaner that sufficiently cleans surfaces to pass ASTM F 22. Sandblast the surfaces with 180-220 grit aluminum oxide. Form a black oxide finish on the surfaces in accordance with MIL-F-495.

30.4 Application on magnesium and magnesium alloys. Preclean the surfaces to be coated with Aliphatic Naphtha conforming to TT-N-95 or any EPA compliant cleaner that sufficiently cleans surfaces to pass ASTM F 22. Anodize the surface in accordance with MIL-M-3171, type I, class A, B, or C.

30.5 Application on steel. Preclean the surfaces to be coated with aliphatic Naptha conforming to TT-N-95 or any EPA compliant cleaner that sufficiently cleans surfaces to pass ASTM F 22 Sandblast the surfaces with 180-220 grit aluminum oxide. Phosphate in accordance with DOD-P-16232 (weight should be 11 - 22 $g/m_2$), type M, class 3 or type Z, class 3.

30.6 Application on stainless steels. Preclean the surfaces to be coated with aliphatic Naptha conforming to TT-N-95 or any EPA compliant cleaner that sufficiently cleans surfaces to pass ASTM F 22. Sandblast the surfaces with 120 grit aluminum oxide. Passivate the surfaces with QQ-P-35, types II, VI, or VII as applicable.

30.7 Application on titanium and titanium alloys. Degrease the surfaces to be coated with aliphatic naphtha meeting the requirements of TT-N-95 or any EPA compliant cleaner that sufficiently cleans surfaces to pass ASTM F 22. Sandblast the surface with 180-220 grit aluminum oxide and alkaline anodize.



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30.8 Engineering tolerances. The operating thickness of this lubricant averages from 0.008 to 0.013 mm per lubricated surface. This thickness seldom requires alteration of established clearances between moving parts. There is one exception. The lubricant coating thickness must be considered in the case of small parts which normally operate with very little clearance. The cured lubricant film is relatively soft and any interference produced by the thickness of the lubricant will cause rapid wear of the lubricant film to the point where interference is eliminated.



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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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| RECOMMEND A CHANGE: | 1 DOCUMENT NUMBER<br>MIL-L-46147B | 2 DOCUMENT DATE (YYMMDD)<br>941202 |
|---|---|---|

3. DOCUMENT TITLE
LUBRICANT, SOLID FILM, AIR CURED (CORROSION INHIBITING)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(if applicable)<br>(2) DSN | 7. DATE SUBMITTED |

8. PREPARING ACTIVITY

| a. NAME<br>Carolyn B. Johnson | b. TELEPHONE (Include Area Code)<br>(1) Commercial (2) DSN |
|---|---|
| c. ADDRESS (Include Zip Code)<br>USA MOBILITY TECH CTR - BELVOIR<br>ATTN AMSTA-RBES<br>10115 GRIDLEY RD STE 128<br>FT BELVOIR VA 22060-5843 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>DEFNS QLTY & STDZN OFC<br>5203 LEESBURG PIKE<br>STE 1403<br>FLS CHURCH VA 22041-3466<br>Telephone (703) 756-2340 DSN 289-2340 |

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